

Hotel Booking Cancellation Analytics: Trends, Predictions and Business Impact

1. Project Overview and Objective

This project involves cleaning, transforming, and analysing raw data using **PYTHON** and creating an interactive **Power BI dashboard** to derive meaningful business insights.

The main objective is to demonstrate data pre-processing techniques using python and an interactive Power BI dashboard visualization to make informed decisions.

2. Data Sources

- **Source Description and Timeline:** Kaggle - Hotel Booking Cancellation Prediction
Dataset import Kagglehub path = Kagglehub.dataset_download("youssefaboelwafa/hotel-booking-cancellation-prediction")
- **Year / Timeline:** Data collected during 2015 to 2017
- **Domain:** Hospitability / Hotel Management etc

3. Problem Statement

- To identify the key factors that influence hotel booking cancellations
- To clean, transform, and preprocess the booking dataset for analysis
- To create meaningful visualizations that clearly explain cancellation patterns and trends
- To provide actionable insights and recommendations to help reduce cancellations

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Booking ID	Categorical (Text)	Unique identifier for each booking
Number of adults / number of children	Numeric (Integer)	Number of adults and children in the booking
Number of weekend nights / number of week nights	Numeric (Integer)	Number of weekend and weekday nights booked

Type of meal	Categorical	Meal plan opted for by the customer
Car parking space	Categorical (Binary)	Whether the customer requested a car parking (0/1)
Room Type	Categorical	Type of room reserved
Lead time	Numerical	Number of days between booking date and arrival date
Arrival year	Categorical/Numerical	Year of arrival
Arrival month	Categorical	Month of arrival
Arrival date	Categorical	Date of the month
Market segment type	Categorical	How the booking was made (Online, Offline, Corporate, etc.)
Repeated guest	Categorical (Binary)	Whether the customer is a repeat guest (0/1)
Average price	Numerical	Average price per day of the reservation
Special requests	Numerical	Number of special requests made by the customer

Booking status	Numerical	Whether the booking was cancelled or not(target variable)
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5. Tools & Technologies

- **Python:** Data cleaning, transformation, and EDA Visualisation.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Python / Power Query)

Tasks Performed:

- **Missing Values & Duplicates:** Verified no missing values across all 17 columns; identified and removed duplicate rows using `drop_duplicates()`.
- **Column Standardization:** Renamed all columns to lowercase with underscores for consistency; dropped `Booking_ID` column as it is a unique identifier not useful for analysis.
- **Outlier Treatment:** Applied the IQR method on `lead_time` and `average_price`; removed 1,332 outlier rows from `lead_time` and 1,608 from `average_price`, reducing the dataset to 33,345 rows.
- **Skewness Check & Transformation:** Checked skewness of numeric columns; applied log transformation (`log1p`) to columns with skewness greater than 1 (e.g., `p-c`, `p-not-c`, `repeated`, `number_of_children`).
- **Data Type Conversion & Feature Engineering:** Converted `date_of_reservation` to datetime format and extracted `reservation_year`, `reservation_month`, `reservation_day`, and `reservation_weekday`; converted `booking_status` into a numeric flag (`booking_status_flag`: 1 = Canceled, 0 = Not_Canceled).
- **Export:** Saved the cleaned dataset as `cleaned_hotel_booking.csv` for use in Power BI dashboard creation.

Exploratory Data Analysis (EDA) Visualizations (Python):

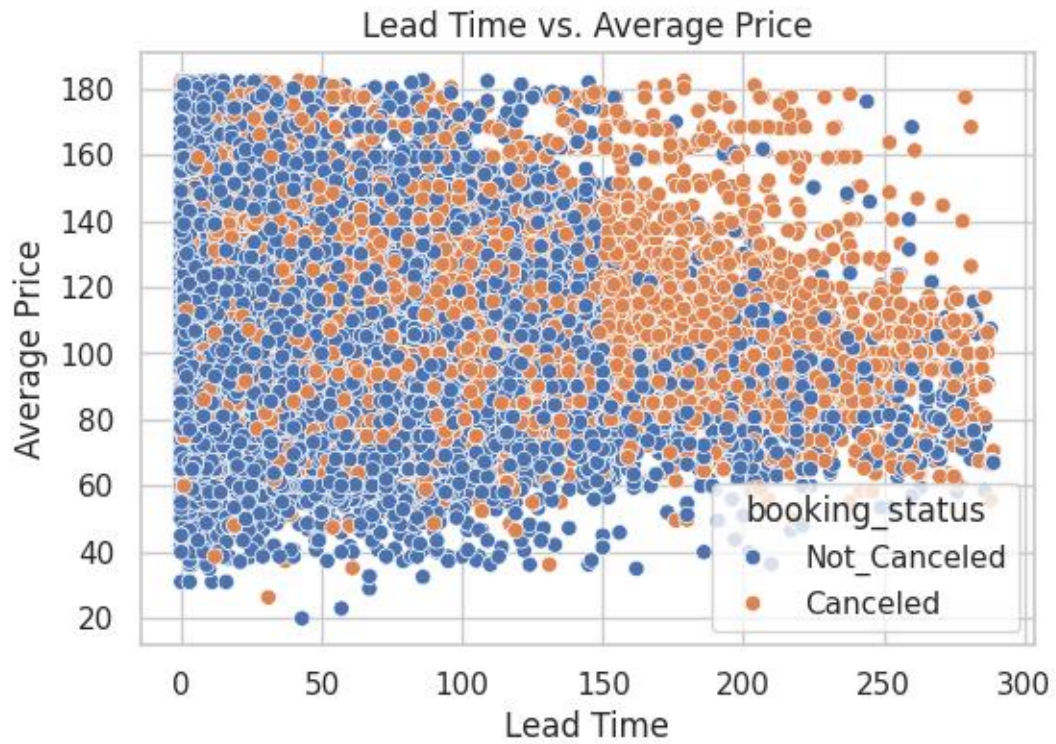


Fig 1: Lead Time vs. Average Price (Scatter Plot)

Cancellations (orange) are noticeably more concentrated among bookings with longer lead times (150+ days), while last-minute bookings (lead time close to 0) are almost entirely Not_Canceled. This confirms lead time as one of the strongest predictors of cancellation.

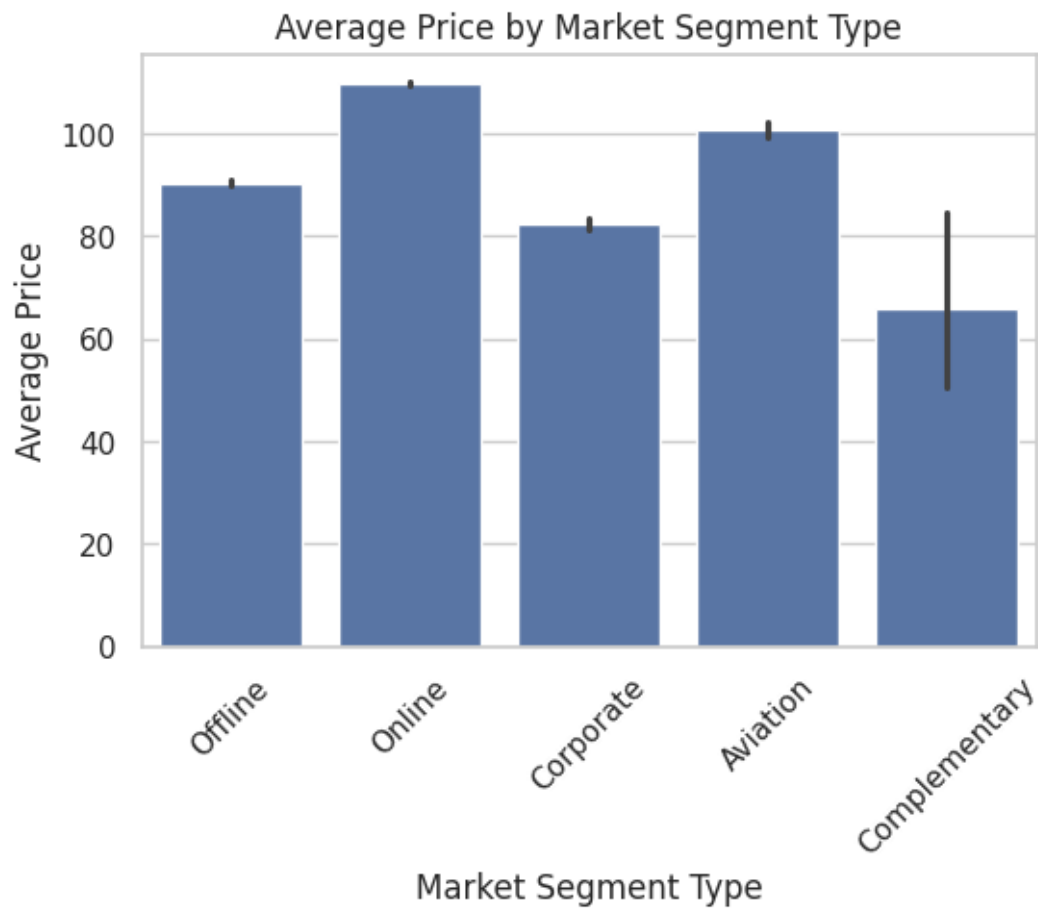


Fig 2: Average Price by Market Segment Type (Bar Chart)

Online bookings have the highest average price (~₹110), followed by Aviation and Offline segments, while Complementary bookings are priced lowest (~₹65) and show the widest variability (largest error bar).



Fig 3: Average Price by Booking Status (Box Plot)

Cancelled bookings have a slightly higher median price (~₹107) compared to Not_Canceled bookings (~₹95), suggesting that higher-priced reservations are marginally more likely to be cancelled, along with a few high-value outliers on both sides.

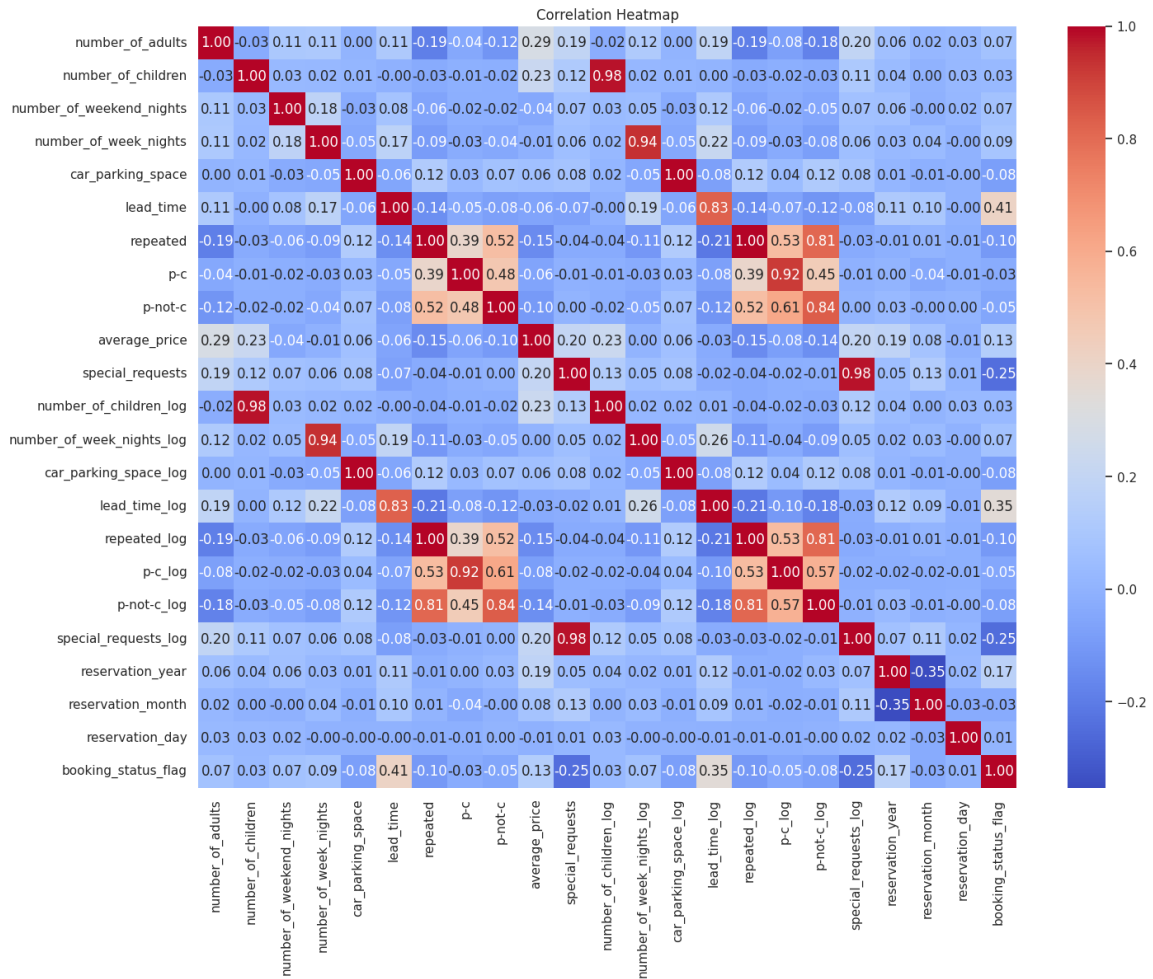


Fig 4: Correlation Heatmap

Lead time shows the strongest positive correlation with booking_status_flag (0.41), reinforcing it as the top cancellation driver, while special_requests is negatively correlated (-0.25), meaning customers who make more special requests are less likely to cancel. Highly correlated pairs like number_of_children vs number_of_children_log (0.98) confirm the log-transformed features preserve the original relationships.

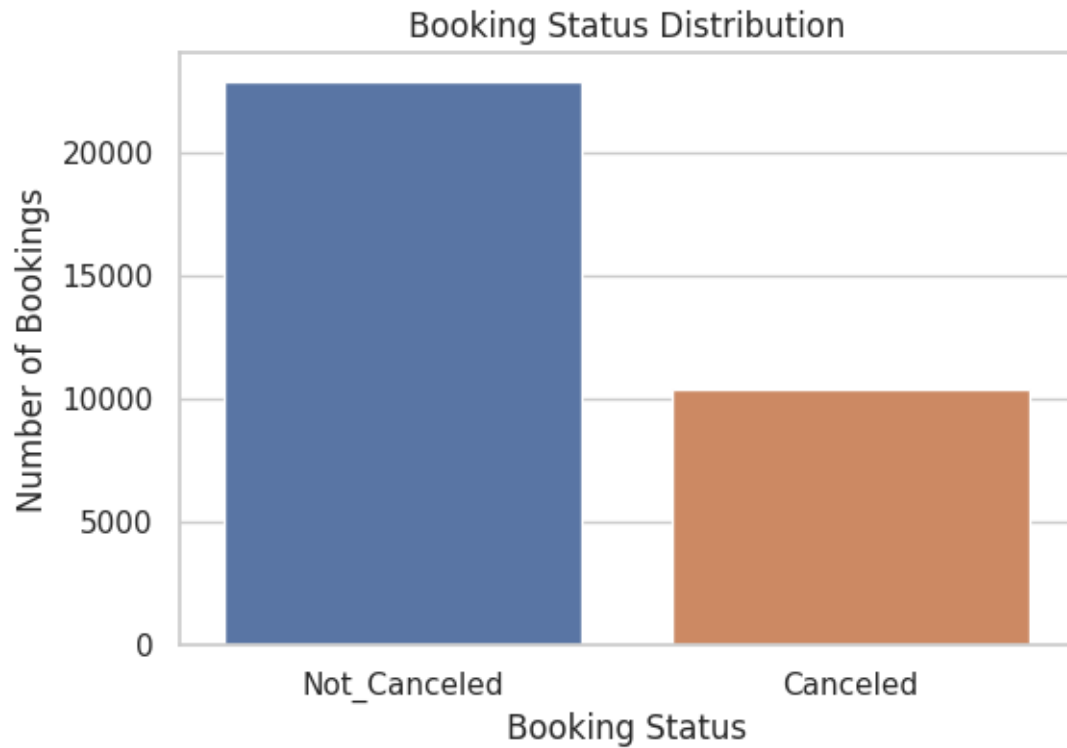


Fig 5: Booking Status Distribution (Bar Chart)

Out of the cleaned dataset, roughly 22,900 bookings were Not_Canceled and about 10,400 were Canceled — a class imbalance (~69% vs 31%) that is important to account for in any predictive modelling.

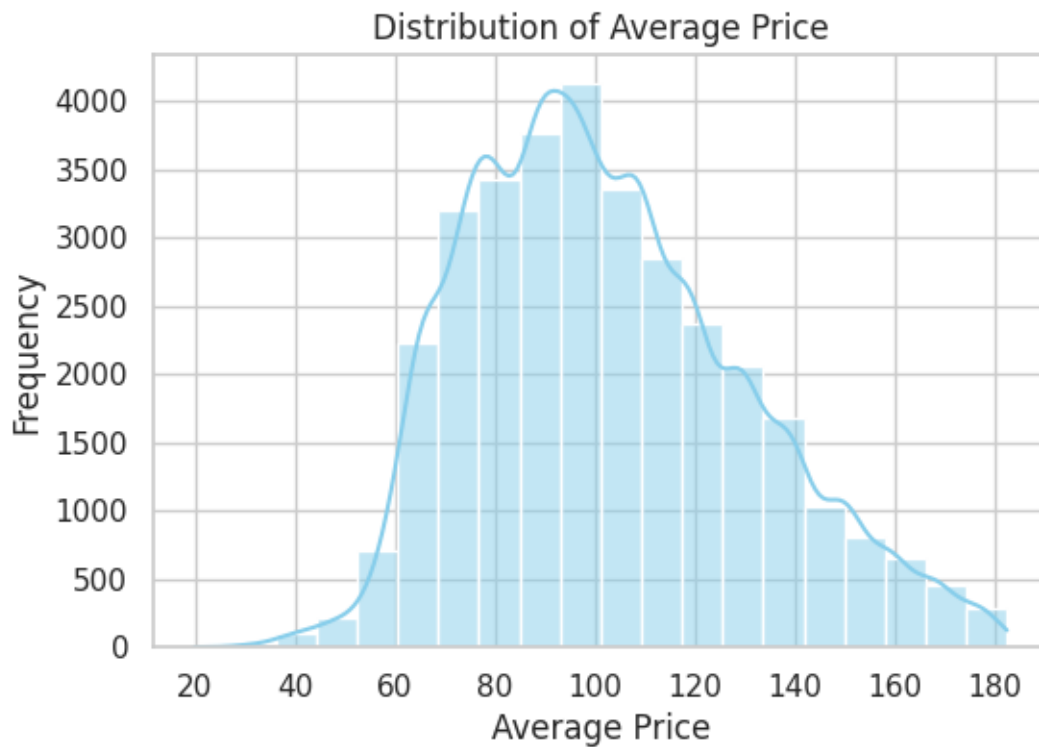


Fig 6: Distribution of Average Price (Histogram)

Average price follows a roughly normal, slightly right-skewed distribution, peaking between ₹80-100 per night, with a long tail of higher-priced bookings extending toward ₹180.

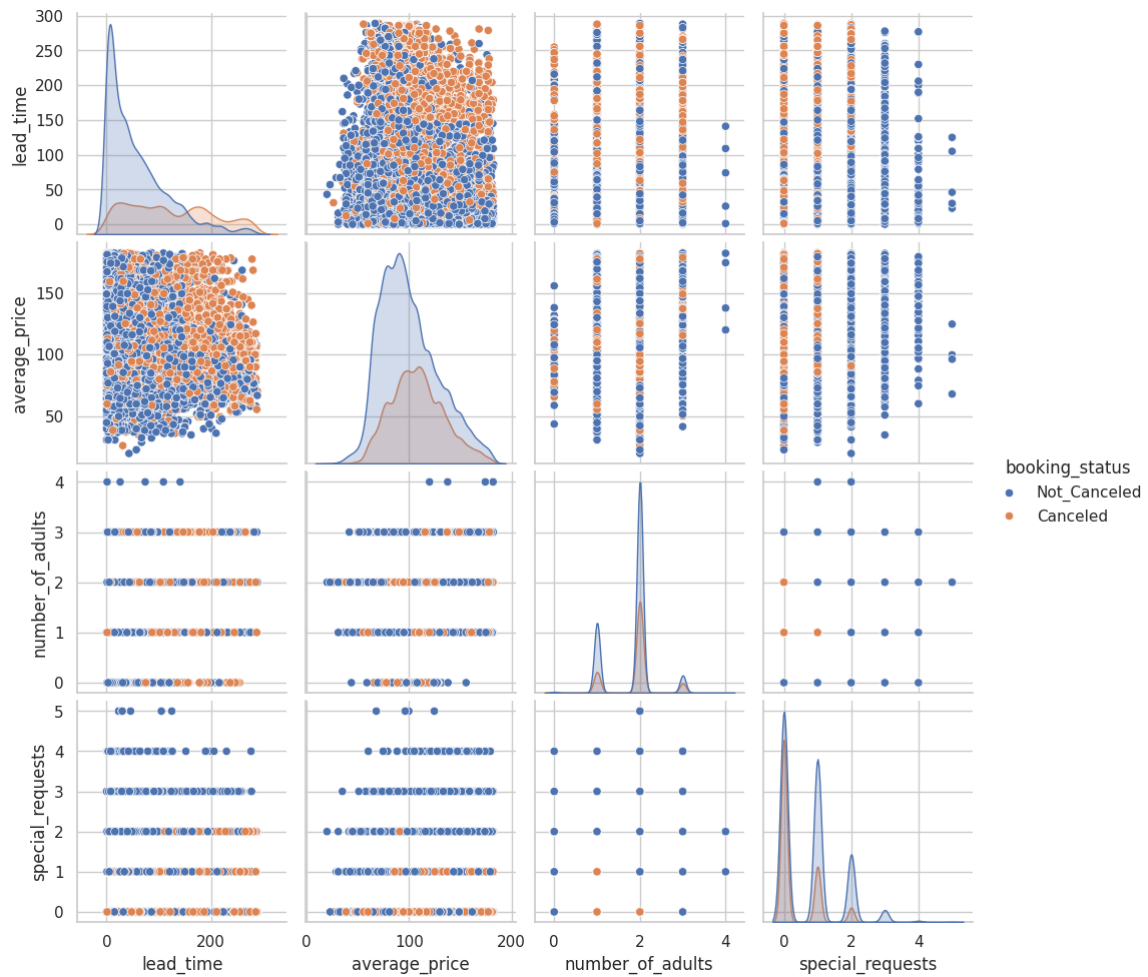


Fig 7: Pair Plot of Key Numeric Features

The pair plot of lead_time, average_price, number_of_adults, and special_requests by booking status shows that cancellations are spread across a wider range of lead times, while non-cancellations cluster more tightly at lower lead times and 2-adult bookings, visually reinforcing the earlier correlation findings.

7. Data Modelling and DAX (Power BI)

- **Data Model:** Since data cleaning, transformation, and feature engineering were already completed in Python, the model uses a single flat table (cleaned_hotel_booking) loaded directly into Power BI. As all required fields exist in one table, no additional relationships or lookup tables were needed.
- **DAX Measures:** Created 9 measures on the cleaned_hotel_booking table to power the dashboard cards and visuals:
 - **Total Bookings** = COUNTROWS(cleaned_hotel_booking)
 - **Total Cancellations** = SUM(cleaned_hotel_booking[booking_status_flag])
 - **Cancellation Rate %** = DIVIDE([Total Cancellations], [Total Bookings], 0) * 100

- **AVERAGE ROOM PRICE** =
AVERAGE(cleaned_hotel_booking[average_price])
- **Avg Week Nights** =
AVERAGE(cleaned_hotel_booking[number_of_week_nights])
- **avg adults** = AVERAGE(cleaned_hotel_booking[number_of_adults])
- **Max Room Price** = MAX(cleaned_hotel_booking[average_price])
- **Total Repeated Guests** = SUM(cleaned_hotel_booking[repeated])
- **Overall Avg Lead Time** =
AVERAGE(cleaned_hotel_booking[lead_time])

8. Analysis and Visualizations (Power BI)

Dashboard Features:

- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, cards, and tables to communicate insights. **(List Out)**
- Make your Report interactive Layout using **Drill-down, Filters and slicers**
- **Bookmarks, Clear Titles for each visuals & Labels is must.**
- **Create a consolidated Report /Dashboard.**

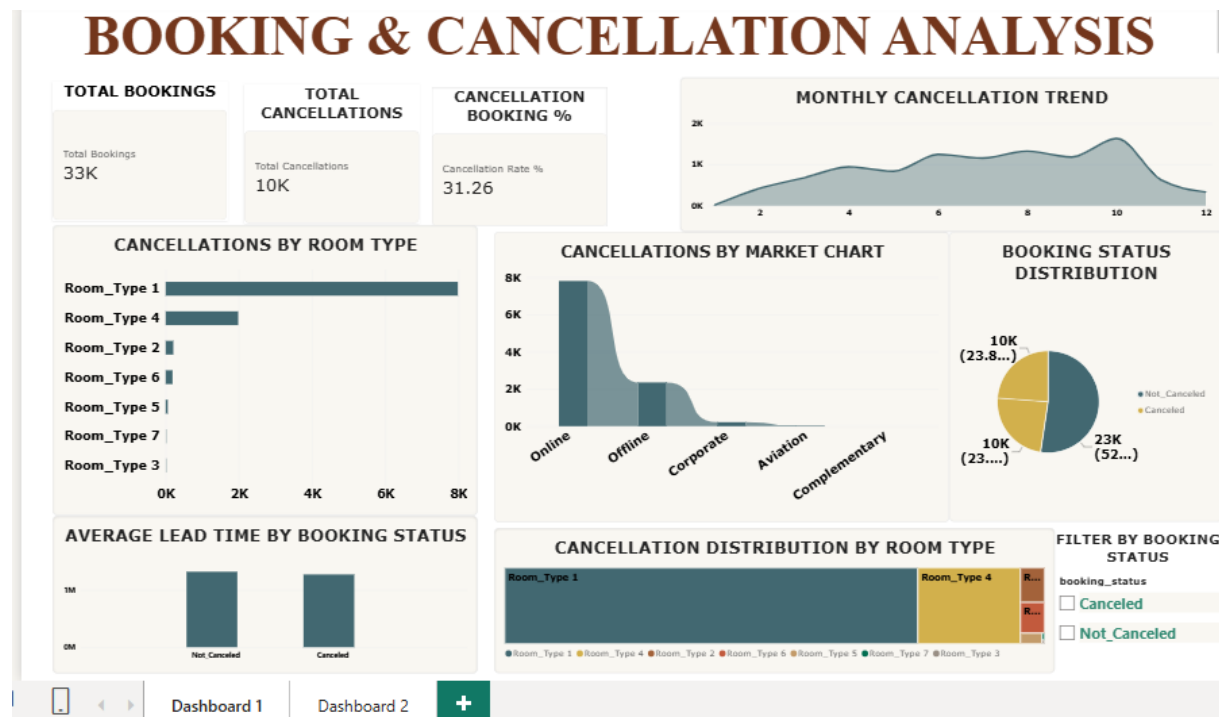


Fig 8: Dashboard 1 — Booking & Cancellation Analysis

Booking & Cancellation Data:

- Total Bookings: 33000 (33k)
- Total Cancellations: 10000 (10K)
- Total Cancellation Rate: 31.26%

- Current page Filter Split: ~52% Not Cancelled vs ~48% Cancelled.

Key Trends:

- **Peak Timing:** Cancellations reach their highest levels during **months 8, 9, 10. (August to October).**
- **Room Type:** **Room_Type1** is responsible for the vast majority of all cancellations.
- **Market Segment:** Online bookings have a noticeably higher average lead time compared to completed stays.

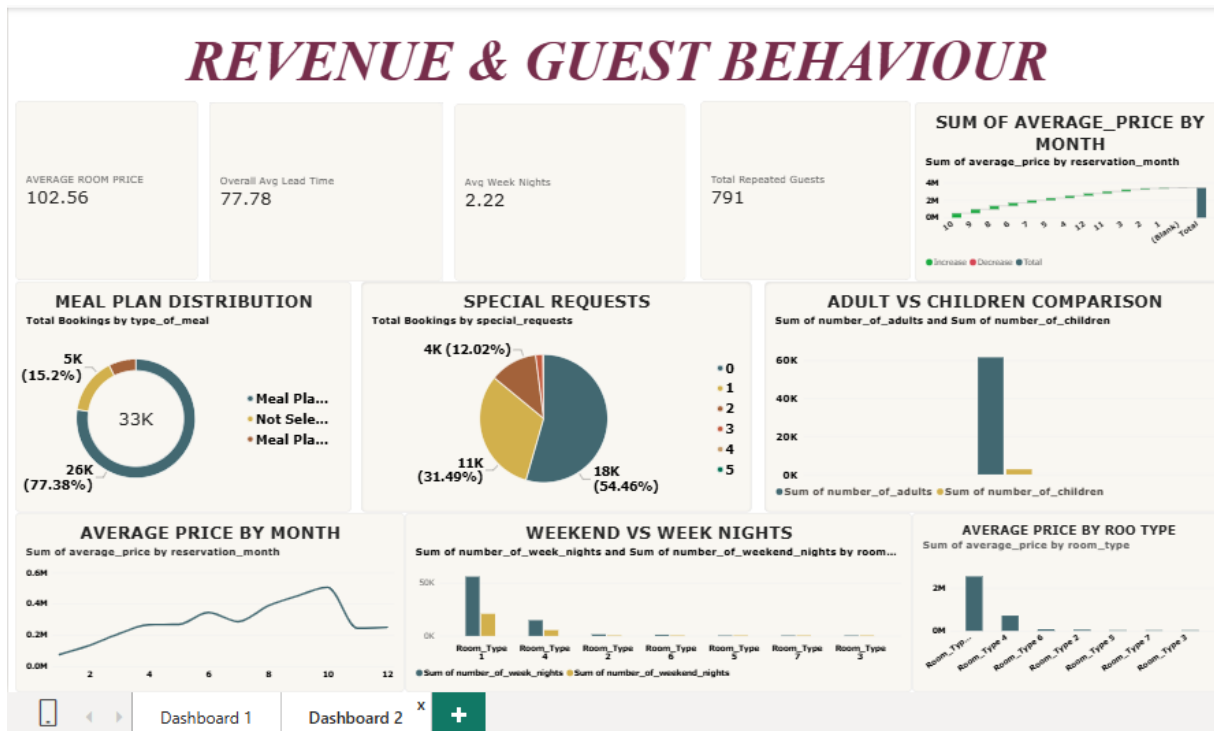


Fig 9: Dashboard 2 — Revenue & Guest Behaviour

Pricing & Rooms:

- Average room price is ₹102.56, which shows an increasing trend throughout the year.
- **Room_Type4** commands the highest average price among all room categories.

Stay Details:

- Average lead time is **77.78 days**, and the average weeknights stay is **2.22 nights**.
- Bookings are overwhelmingly made by adults rather than children.

Preferences & Loyalty:

- **Meal Plan 1** is heavily preferred, dominating **77.38%** of all bookings.
- Over half of the bookings (**54.46%**) come with **zero special requests**.
- There are **791 repeated guests** who are returning customers.

9. Insights & Conclusions

- **Key Findings:**
 - Total Cancellations: Out of 33345 cleaned bookings, 31.26% were cancelled.
- The Lead Time Effect:
 - The longer the time between booking and arrival, the higher the chance cancellation.
 - Once the booking lead time crosses ~150 days, cancellations rise sharply.
 - In contrast, last-minute bookings almost never cancel.
- Special Requests:
 - Guests who make special requests (like an extra bed or higher floor) **are less likely to cancel.**
- Higher Room Prices:
 - Cancelled bookings tend to have a higher average room price (**~₹107 Median**) compared to completed stays (**~₹95 median**).
- **The Main Culprits:**
 - The combination of **Room_Type1** and the **online market segment** accounts for the vast majority of all cancellations.
- **Low Guest Engagement:**
- Only 791 out of 33k are repeat guests (very rare).
- 54.46% of bookings come with zero special requests.
- Both are red flags showing low customer connection, which leads to
- higher cancellation risks.
- **Provide the analysis insights:**
 - **Descriptive** (What happened):
 - Processed 33,345 total bookings with an average room price of \$ 102.56.
 - Recorded an average booking lead time of 77.78 days across all segments.
 - Registered an overall cancellation rate of 31.26%.
 - Identified `room_type1` and the online distribution channel as the primary contributors to cancellations.

- Observed peak cancellation trends specifically during months 8-10.
- **Diagnosis** (Why it happened):
 - Long lead-time bookings give guests more time to find better deals or change plans.
 - Online channel bookings lack upfront financial commitment or deposits, making them highly friction free to cancel compared to corporate or aviation segments.
 - Guests adding special requests (extra bed, high floor) show a stronger personal commitment to their stay, leading to a negative correlation with cancellation rates.
 - Low guest engagement is heavily driven by a lack of repeat visitors, with only 791 out of 33k being loyal guests.
- **Predictive** (What is likely to happen):
 - Online bookings with >150 days lead time and prices above \$ 100 will continue to cancel the most.
 - Bookings without any special requests will remain a high risk for revenue.
 - Occupancy and revenue will drop every year during months 8-10 if left unmanaged.
- **Prescriptive** (What should be done):
 - Ask for advance deposits or make bookings non-refundable if the lead time is over 150 days.
 - Offer special discounts or packages to attract guests during the low months (8-10).
 - Add simple checkboxes for special requests during online checkout to increase guest commitment.
 - Start a loyalty program to convert one-time online bookers into repeat guests.

Project Scope: Successfully combined Python and Power BI to analyze 33,345 total hotel bookings.

Core Focus: Evaluated key factors like room choice, market trends, cancel status and guest behaviour.

Major Findings: Long booking lead times and high room choice, market trends, cancel status, and guest behaviour

Engagement Insights: A large part of the data highlights a clear drop in customer interaction and low repeat guest retention.

The Solution: Implementing non-refundable deposit rules and active loyalty outreach will protect revenue and stop future booking losses.